

End of year report

Plasma processing of PAHs

and related molecules –

Theoretical and

experimental approaches

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Abstract

In the context of the energetic processing of Polycyclic Aromatic Hydrocarbons (PAHs), understanding their reaction mechanisms under plasma conditions is crucial to unravel the various roles these molecules play in the chemistry and physics of the interstellar medium (ISM). Computational tools, and in particular, plasma kinetic modelling constitute a very suitable framework to address this challenge. In this work, a zero-dimensional model of a dielectric barrier discharge (DBD) reactor has been developed, providing a novel and simplistic approach for including experimental parameters as discharge power and flow rate, as well as focusing on a drastic reduction of the computational cost. This model has been tested with a simple molecule as CO₂ and validated against experimental data showing a very good quantitative agreement. Finally, ongoing experiments with Naphthalene PAH model compound in a low pressure radio frequency plasma are described.

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1. Introduction

The EUROPAH European Training Network aims to study the extensive and ubiquitous role of Polycyclic Aromatic Hydrocarbons (PAHs) in space. In the context of the energetic processing of PAHs, the present PhD project focuses on their interaction with energetic electrons, i.e. plasma processing of PAHs. The main objective is to identify fragmentation patterns under plasma conditions so as to understand reaction mechanisms for PAHs' decomposition in the interstellar medium (ISM), both from a theoretical and experimental approaches.

The theoretical part of this project consists on the development of a plasma kinetic model for PAHs in different reaction setups like dielectric barrier discharge (DBD) or gliding arc (GA) reactors. Due to its chemical complexity, such model is kept as a long term objective since important input information as electron impact cross sections and reaction rate coefficients is not yet available. So far, a model for a DBD reactor has been developed and tested with a much simpler molecule (CO_2) showing satisfactory results. This molecule was chosen on the basis of its relevance for the energy and environmental fields. Moreover, experimental data of CO_2 dissociation in DBD reactors was readily available within Dr. Tu's group, which was used to validate the model's predictions [1].

Besides, the experimental part of this project involves plasma processing of Naphthalene (as the simplest PAH model compound) in a radio-frequency low pressure plasma chamber. The aim of these experiments is to identify fragmentation patterns of Naphthalene under plasma conditions by means of mass spectrometry. Preliminary tests on this experimental setup have been recently conducted.

2. Literature review

2.1. Plasma kinetic modelling

Computational tools are deemed crucial to understand the underlying chemistry of a given system or reaction. In particular, plasma kinetic modelling represents a suitable approach for tackling this issue and provides a great tool to analyse the relative relevance of various processes taking part in a reaction mechanism [2]-[8].

Plasma discharges in DBD reactors are normally modelled through zero-dimensional models. Within this framework, spatial variations of any property in the radial and angular coordinates of a tubular reactor are neglected while the remaining dimension (axial coordinate) is translated into time by means of an average fluid velocity [2]-[8]. Then, each element of volume flowing through the reactor is treated as a small batch reactor progressing in time until reaching the residence time of the reactor.

Two- and three-dimensional models of plasma systems have also been developed [9][10], nevertheless zero-dimensional models have been applied satisfactorily to DBD reactors [2]-[8]. Moreover, the latter models are much simpler than dimensional ones and represent a smaller computational cost.

In these models, the number densities of each species are calculated as a result of production and loss processes, i.e., chemical reactions. Among the chemical systems modelled with this approach, CO₂ splitting [5][6][7], CO₂ with additional reactants [8], Methane conversion [4] and dry reforming [2][11] can be found in the literature. The number of chemical reactions included in these models can reach several hundreds and therefore represent a trade-off between accuracy and computational cost. In the case of an extensively studied reaction as CO₂ dissociation, a reduced set of 17 chemical reactions out of a total 501 reactions has been reported to accurately represent the chemistry of the system for low conversions. Discrepancies between these two models arise at higher conversions, as products reach higher concentrations and the chemistry of the system becomes more complex [5].

Plasma discharges in DBD reactors exhibit a filamentary regime for most gases [12]. In order to mimic such condition, these spatially spread filaments can be represented as consecutive discharges or pulses over time with a certain frequency of pulses. Importantly, this frequency has been varied throughout a wide range of values by different authors regardless of the physical phenomena involved. The frequency of the applied voltage has been used as the frequency of pulses (for instance, 70 kHz in [3]), even though they are not strongly related. Moreover, some arbitrarily chosen values have also been used, ranging from 1000 kHz [2] to 0.03 kHz [5]. The physical meaning of this parameter will be discussed in this work (see section 3.1).

2.2. Modelling of CO₂ conversion reactions

The conversion of CO₂ into value added chemicals or fuels has attracted much attention over the past years, both because of its high availability as an exhaust gas and its negative environmental impact in terms of greenhouse effect and global warming. Unfortunately, CO₂ is a rather stable molecule and high conversions are not easy to achieve without a fairly poor energy efficiency. Many research efforts have been destined to overcome these difficulties, among which plasma processing is presented as a promising alternative due to their non-equilibrium nature, deemed crucial for CO₂ dissociation reactions. Gaining a deep understanding of the reaction mechanisms involved under plasma conditions is essential to improve the performance of such process.

As reported elsewhere, CO₂ conversion is believed to proceed through three main mechanisms, namely electron attachment, electron impact dissociation and electron impact ionisation (accounting

for 11, 46 and 43% of the total conversion, respectively) [5]. However, it was later realised that these relative contributions depended heavily on the cross section data used to model the electron impact interactions, being this input crucial for the validity of the model [6]. In this regard, it is very important that, whenever possible, complete and consistent sets of cross sections (i.e. validated against swarm parameters) are used [13].

Particularly, there is no clear consensus about the most suitable cross section for the dissociation of CO₂ by electron impact excitation, which is importantly one of the main mechanisms for this reaction. This excitation process has been extensively studied in [13] and [6], and numerous cross section data sources are available for it in the LXCat database [14]. Mainly three data sets have been used for this interaction, namely Phelps' [15], Itikawa's [16], and Polak and Slovetsky's [17]. The first set is likely to involve more processes than only dissociation, overestimating the rate coefficients as explained in [13]. Itikawa's set, used in [2]-[5], has been recently reported to underestimate the CO₂ dissociation reaction rate coefficients [6][13]. Besides, Polak's cross sections are believed to represent CO₂ dissociation quite accurately [13]. Nevertheless, different authors recommend different datasets to model CO₂ dissociation (Phelps's 7 eV or Polak's datasets according to the results reported in [6] and [13], respectively). In this work, the latter recommendation was followed [13].

Another ongoing discussion relates to the energy efficiency of CO₂ dissociation reactions and the role that vibrational excitations play in it. This topic will not be addressed in this report as CO₂ splitting is not the main subject of it, while the reader is referred to ANNEX I – 3.5 for this interesting discussion. The main conclusion indicates that vibrationally excited states are not an important channel for CO₂ dissociation in DBD reactors while they could be responsible for their low efficiency.

2.3. PAHs in the ISM

Large PAHs carry the infrared (IR) emission features that dominate the spectra of almost all interstellar objects. Then, interstellar PAHs are ubiquitous and lock up around 10% of the elemental carbon in the ISM [18][19]. Besides the strong evidence for their presence in space, their numerous roles in chemistry and physics of the ISM are yet to be fully understood. It is believed that PAHs are destroyed by energetic photons, ions or electrons. Also, given that PAHs offer a great surface area available for catalytic reactions, they may act as catalysts for the formation of H₂ (the most abundant molecule in the ISM). Although some experimental works tend to confirm these features of PAHs in the ISM [20]-[22], the details and mechanisms of these processes and interactions are poorly understood.

A number of key differences between the ISM and terrestrial environments shall be taken into account when trying to study the role of PAHs in the ISM, meaning that experiments under terrestrial conditions can only answer these questions to a certain extent. Conditions in space are extreme with very low pressures, consisting of low level plasmas with low ion fractions and weak radiation fields (but energetic photons). Very large timescales are involved in the chemistry of the ISM, from one day to hundreds of millions of years. Then, processes with very low probability can be very important over the lifetime in space. In the quest to recreate the ISM environment, experiments under low pressure or ultra-high vacuum are commonly used.

Interstellar PAHs are processed by energetic UV photons and ions in the ISM, affecting mainly less stable species [23]. As a result, the local physical conditions cause changes in the composition of the interstellar PAH family. Such changes can be inferred from the variation that the IR spectral signatures of PAHs show between different sources [24]. Besides, interactions with energetic electrons may also be relevant for the chemistry of PAHs in the ISM, being particularly important to study regions with a low radiative field and also to understand the synergistic effects of plasma and UV radiation.

Regarding plasmas, the fragmentation patterns of PAHs and the products formed depend on the electron energy distribution and, ultimately, the plasma processing parameters which induce different physical and chemical conditions. However, there is a lack of experimental and theoretical data about plasma processing of PAHs so as to understand the various plasma-induced physical and chemical processes that may occur in the ISM.

3. Plasma kinetic modelling

3.1. Description of the model

3.1.1. Experimental setup to be modelled

A co-axial DBD reactor with no packing and a discharge gap of 2.5 mm will be modelled in this work (see Figure 1 for dimensions). Different values for the discharge power and flow rate are considered (10 to 50 W and 25 to 125 mL/min, respectively), corresponding to different experimental measurements. Further details on the experimental setup can be found in [1].

3.1.1. Zero-dimensional model

Starting from the zero-dimensional models described in section 2.1, our focus was on understanding several processes taking place at different time/space scales. These scales range from nanoseconds

(duration of free electrons in a plasma filament) to microseconds (repetition of the pulses) and milliseconds (time it takes a molecule to go through a filament). Also, the time between filaments is deemed to be one order of magnitude higher than the latter.

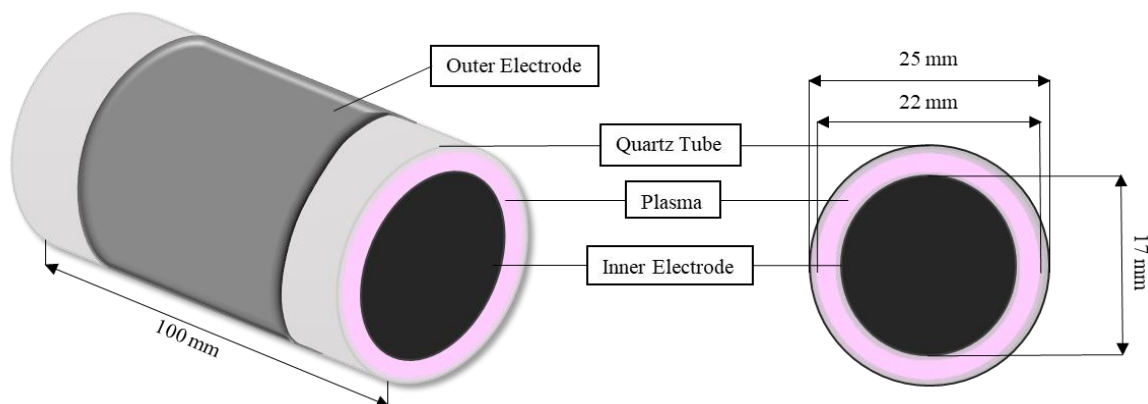


Figure 1 – Reactor Scheme

The frequency of pulses arises from this analysis as the ratio between the fluid velocity and the spacing between filaments. The latter is not known, but an order of magnitude of 1 mm can be assigned based on experimental observations. Slight variations of this parameter around 1 mm were proposed as a function of the discharge power, based on experimental observations indicating an increasing number of filaments per unit area with increasing discharge power [25]. This resulted in a frequency of pulses ranging from 0.002 to 0.01 kHz for the different operating conditions. This range is much narrower than the spread found in different publications (from 0.03 to 1000 kHz) and is based on the flow phenomena taking place inside the reactor.

In a different scale, the time spent by a molecule inside a filament was calculated to be in the order of the milliseconds according to the filament width reported in [12]. At this scale, a higher flow rate (or velocity) implies a lower time inside the filaments (where electron impact reactions take place). Since the interaction with electrons is what really drives the dissociation of CO₂, a higher velocity causes a drop in the conversion. Nonetheless, we also propose that a higher flow rate affects the so called *memory effect* [12] of DBD reactors. In our model, a higher velocity and turbulence spreads the micro-discharge remnant widening the spot where the next micro-discharge could occur, as well as the filament itself. This has a slight positive effect in the conversion with increasing flow rate which balances the described drop in the conversion.

Finally, the phenomena taking place at the nanosecond and microsecond scales (a micro-discharge and its repetition, respectively) could hardly be modelled at a reasonable computational cost. For this

reason, the hundreds of micro-discharges taking place when the element of volume passes through a filament were averaged over the period of time inside that filament, in the order of the milliseconds. In this way, we obtained an average electron density of $4 \times 10^9 \text{ cm}^{-3}$ valid for the whole period inside the filament, calculated from a real electron density of $1 \times 10^{13} \text{ cm}^{-3}$ in each 30 ns-long micro-discharge. This electron density and micro-discharge duration are in agreement with the values used in several works and reported in [12]. The aim of this approximation was to achieve a sharp reduction in the computational time required to complete these calculations. The analysis of the different scales involved in this model are illustrated in ANNEX I – Graphical abstract.

3.1.2. Chemical model for CO₂ splitting

In this work, a reduced chemistry set for CO₂ conversion already developed and reported in [5] was adopted. This reduced set consists of 9 species and 17 reactions, which implies a much lower computational cost than the complete set (around 42 species and 500 reactions) [2],[5]. The reactions considered can be found in ANNEX I – Table 2. As regards CO₂ conversion by electron impact reactions, the main mechanisms identified in [2] are taken into account, i.e., electron attachment (R1), electron impact dissociation (R2) and total ionisation (R3). Vibrationally excited states of CO₂ were not taken into account in this model (see [2], [7] and ANNEX I – 3.5).

The time integration of the production and loss processes of the different species was done by means of the Fortran module ZDPlasKin [26], implemented by a user code developed as part of this work. In this way, the number densities of the different species were calculated as a function of time. The rate coefficients of reactions between heavy molecules were taken from the literature (see ANNEX I – Table 2 for detailed references). The rate coefficients for electron impact reactions were calculated by the Boltzmann solver BOLSIG+ [27], coupled with ZDPlasKin, as a function of the corresponding cross sections. Cross section data was retrieved from different data bases within the LXCat project [14]. Polak and Slovetsky's cross section for CO₂ electron impact dissociation has been used [13][17].

3.2. Results and discussion

3.2.1. Validation of the model – Average Electron Density

The averaging process described in section 3.1.1 has been verified by comparing the calculations of an averaged pulse and a detailed one, consisting of 662 micro-discharges. The results in conversion are 0.30286 % and 0.30277 %, respectively. The absolute error is 0.0001 % while the relative error equals 0.0003 (0.03%). Moreover, the detailed calculation ran for 450 minutes while it took less than a minute to complete the averaged calculation, resulting in a time saving of more than 99%.

Furthermore, it would take months to run the detailed calculation over a residence time of the reactor while the averaged calculation typically finishes within one or two hours. Thus, we consider this approximation not only reliable but also necessary.

3.2.2. Validation of the model – Matching experimental data

The results predicted by the model were contrasted against experimental data taken from [1]. The calculated CO₂ conversion figures are in very good agreement with the laboratory measurements, with only one point having a relative error above 5%. The complete set of results including every calculation is presented in Table 1 and displayed graphically in Figure 2 as a function of SEI (specific energy input, defined as the ratio between the power and the flow rate).

Table 1 - Results. All calculations with gap = 2.5 mm and length = 100mm.

Flow rate mL/min	Power W	SEI kJ/L	EXP. CO ₂ Conversion %	MODEL CO ₂ Conversion %	Relative error %	EXP. Efficiency %	MODEL Efficiency %
25.0	10	24	17.4	17.4	0.2	8.3	8.3
25.0	20	48	19.8	19.6	1.1	4.7	4.7
25.0	30	72	21.0	20.9	0.3	3.3	3.3
25.0	40	96	22.0	22.0	0.2	2.6	2.6
25.0	50	120	22.4	22.7	1.3	2.1	2.2
25.0	50	120	22.4	22.7	1.3	2.1	2.2
31.2	50	96	20.8	20.9	0.3	2.5	2.5
41.2	50	73	18.0	18.8	4.7	2.8	2.9
62.5	50	48	15.8	16.0	1.5	3.7	3.8
125.0	50	24	12.6	12.0	5.5	6.0	5.7

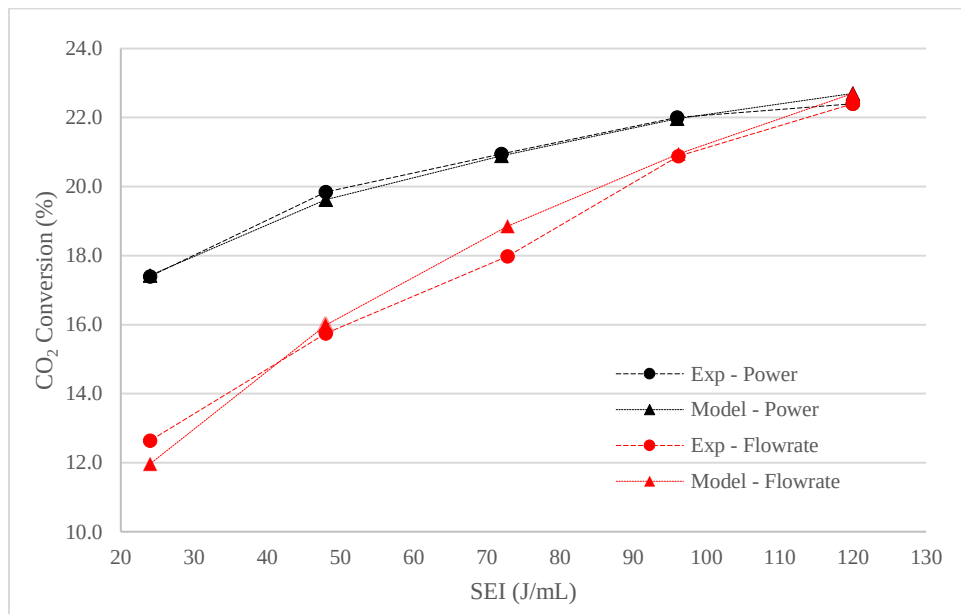


Figure 2 – CO₂ conversion at different discharge powers and flow rates in terms of SEI.

It can be seen that the CO₂ conversion increases with increasing discharge power and decreasing flow rates. These trends, already identified in [1], could be replicated not only qualitatively but also quantitatively by our model. Once this agreement is achieved, the model is considered reliable enough to analyse the reaction mechanisms taking place in the system under study.

3.2.3. Analysis – Reaction mechanisms

Starting from CO₂ with a number density of $2.45 \times 10^{19} \text{ cm}^{-3}$, these figures drop to $1.89 \times 10^{19} \text{ cm}^{-3}$ for the highest conversion achieved (22.7%), being CO and O₂ the main products formed ($5.56 \times 10^{18} \text{ cm}^{-3}$ and $2.77 \times 10^{18} \text{ cm}^{-3}$, respectively). In addition, the major by-product is O₃, although several orders of magnitude below the O₂ final density.

From the calculated reaction rates, it can be concluded that the electron impact dissociation is the dominant mechanism for CO₂ splitting accounting for around 95% of the conversion. The electron attachment represents the remaining 5% while the contribution of the total ionisation to the CO₂ conversion is negligible. These results agree qualitatively with [6] and [13]. Previous DBD models predicted a more balanced contribution of these processes as a result of the cross sections used [2]-[5].

Even though the CO produced undergoes some recombination reactions, these are minor and this product is rather stable. By contrast, the O radicals produced react mainly towards O₂ formation rather than recombining with CO molecules towards CO₂. See ANNEX I – Fig. 8 for an illustration of the main reaction mechanisms within this chemistry set.

4. Experimental part

4.1.1. Experimental Setup

A low pressure – radio frequency (RF) rig is being used as the experimental setup for this study. The vacuum chamber consists of a cross-tube glass reactor with a copper coil around it (not in contact with the plasma). RF power is applied to the coil by means of a plasma source of 13.6 MHz, also connected to a manual impedance matching unit. In this way, a RF inductively coupled plasma is generated in the vacuum chamber. A turbo-molecular pump backed by a rotary pump sets the base pressure below 10^{-5} Torr.

To start with, Naphthalene is used as the simplest PAH model compound. This reactant is presented as a white or colourless powder. A relatively small amount of this compound (around 0.2 g) is placed in a glass bulb connected to the chamber. The lid of this bulb has a screw that acts as a valve between

the bulb and the chamber. When this valve is partially open, the pressure inside the bulb drops up to a point where Naphthalene vapours are generated by sublimation and leaked into the vacuum chamber. In this way, there is no need to enhance vapour release by heating Naphthalene and the powder in the bulb remains at room temperature throughout the whole experiment.

A Hiden EQP quadrupole mass spectrometer is attached to the chamber and will be used to analyse the species formed in the plasma. When Naphthalene is leaked to the chamber, these measurements will allow to identify the fragmentation patterns of this PAH model compound under plasma conditions.

4.1.2. Ongoing works

By the time being, only preliminary tests have been done in this rig. The system has been satisfactorily tested for leaks, reaching a base pressure of 6.0×10^{-6} Torr. In addition, the strategy to feed Naphthalene into the chamber was put in practice to address whether the pressure inside the bulb was low enough to sublime Naphthalene and also to make sure that the pressure drop across the connection was not high enough to stop the vapour generation (in which case the system would be unstable).

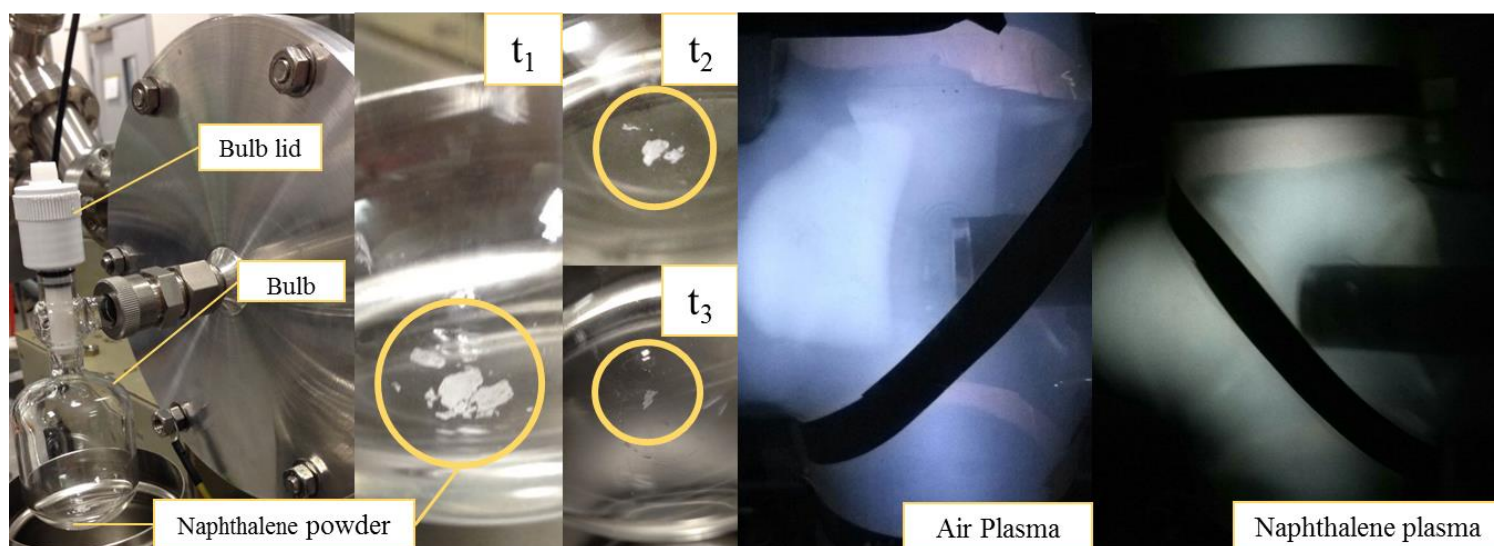


Figure 3 - Naphthalene in bulb (left) and different colours of plasmas (right)

Upon opening of the bulb lid, the pressure stabilized for several minutes in a value one or two orders of magnitude higher than the base pressure (depending on the opening). Then, the base pressure was recovered when the lid was closed. At this point, the possibility of air leaking through the bulb lid when it is partially open to let Naphthalene in has not been ruled out. Nevertheless, the powder inside to bulb showed signs of being consumed during the experiment (see Figure 3 - left), indicating that

either Naphthalene alone or a Naphthalene-air mixture was fed to the chamber when the lid is open. Mass spectrometry measurements will easily confirm this. In addition, we identified a clear difference in the colour of the plasma generated with residual Air only and during the Naphthalene experiments, indicating a different plasma composition (see Figure 3 - right).

5. Conclusions

In this work, a simplistic approach was proposed to develop a zero-dimensional plasma kinetic model for a DBD reactor aiming at a sharp reduction of the computational cost. A description of the reasoning behind the model choices is presented, explaining the dependencies proposed between the model parameters and the power and flow rate values. Due to the complexity of a chemical kinetic model for PAHs, our DBD model has only been tested with CO₂ dissociation reactions at this point. Overall, the predictions of the model presented a very good agreement with the available experimental data within the whole range of operational parameters tested, being the maximum relative error of around 5.5%. The main channel for CO₂ splitting was found to be the electron impact dissociation. Besides, experiments with Naphthalene as the simplest PAH model compound have been preliminary conducted. The experimental setup has been recently tested and a Naphthalene containing plasma was formed.

6. Output

A manuscript titled “A simplistic approach to plasma kinetic modelling of CO₂ conversion in a dielectric barrier discharge reactor” has been submitted to the journal Plasma Sources Science and Technology. Many of its findings and conclusions have not been included in this report as CO₂ splitting is not its main subject (and also due to the 10-page limit of this report). Nevertheless, the manuscript is included as ANNEX I and the reader is referred to it for a deeper discussion on the topics covered by sections 2.1, 2.2 and 3.

An abstract titled “Plasma reforming of Naphthalene as biomass tar model compound for syngas production” has been submitted to the 22nd International Symposium on Analytical and Applied Pyrolysis (Kyoto, Japan June 2018)

7. Training report

Among the several trainings attended during this PhD program, two were taken as part of the EUROPAH project, namely:

- EUROPAH 1st Training Event (Aarhus Universitet, Denmark, September 2017) – PAHs in the ISM, public engagement and surface science workshop.
- PAH in the ISM: Experimental, Observational and Computational tools (Ecole de Physique, Les Houches, France, April 2018).

8. Future plans

The targets for the upcoming period are as follows:

- Plasma kinetic modelling (May 2018 - May 2019): Development of a chemistry set for Naphthalene as a PAH model compound, to be implemented in the existing DBD plasma kinetic model. This includes a thorough literature review and possibly further calculations to approximate the electron impact cross section data set for Naphthalene. Also, the reactions between heavy molecules and plasma species must be addressed and included in the model.
- Ongoing experiments (May 2018 – Sept. 2018): Firstly, it is necessary obtain mass spectrometry measurements of the Naphthalene plasma described in section 4.1.2. This could give us some insight on the decomposition of PAHs in the absence of hydrogenation/oxidation reactions. Furthermore, several other alternatives in terms of plasma-forming gases are worth exploring to allow comparison and address the interactions of PAHs with different plasma species. For instance, a Hydrogen plasma would give an important insight regarding hydrogenation reactions while an Air plasma (with water content) could be useful to study reactions with O and OH radicals.
- Output (Oct 2018 – Nov 2018): A paper will be written with the results obtained from the described experiments. In addition, a deliverable for the EUROPAH ETN will be presented, gathering all our results to date.

9. References

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